

$G(x, y)$ :  $x$  has received all  $y$  grades.

$$\exists x (C(x, 4) \wedge G(x, A))$$

Exercises 40-44 deal with the translation between system specification and logical expressions involving quantifiers.

40. Translate these system specifications into English, where the predicate  $S(x, y)$  is " $x$  is in state  $y$ " and where the domain for  $x$  and  $y$  consists of all systems and all possible states, respectively.

a)  $\exists x S(x, \text{open})$ : There exists a system which is open.

b)  $\forall x (S(x, \text{malfunctioning}) \vee S(x, \text{diagnostic}))$ : Every system is either malfunctioning or is diagnostic.

c)  $\exists x S(x, \text{open}) \vee \exists x S(x, \text{diagnostic})$ : There exists a system which is open or there exists a system which is diagnostic.

d)  $\exists x \neg S(x, \text{available})$ : There exists a system which is not available.

e)  $\forall x \neg S(x, \text{working})$ : No system is working.

41. Translate these specifications into English, where  $F(p)$  is "Printer  $p$  is out of service,"  $B(p)$  is "Printer  $p$  is busy,"  $L(j)$  is "Print job  $j$  is lost," and  $Q(j)$  is "Print job  $j$  is queued."

a)  $\exists p (F(p) \wedge B(p)) \rightarrow \exists j L(j)$ : If there exists a printer which is out of service and is also busy, then there exists a print job which is lost.

b)  $\forall p B(p) \rightarrow \exists j Q(j)$ : If all the printers are busy, then there exists a print job which is queued.

c)  $\exists j (Q(j) \wedge L(j)) \rightarrow \exists p F(p)$ : If there exists a print job which is queued and is also lost, then there exists a printer which is out of service.

d)  $(\forall p B(p) \wedge \forall j Q(j)) \rightarrow \exists j L(j)$ : If all the printers are busy and also all the print jobs are queued, then there exists a print job which is lost.